

SHELL PROGRAMMING PRACTICAL

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File: script.sh

AIM:

To create a shell script using vi editor, make it executable using chmod, and perform various operations like file handling, loops, and logical tasks.

ALGORITHM:

1. Open terminal
2. Create a shell script using vi editor
3. Write commands in the script
4. Display current working directory
5. List files in the directory
6. Create a folder and file
7. Accept file name and count words
8. Reverse a string
9. Check even or odd number
10. Find largest of three numbers
11. Print multiplication table
12. Check palindrome
13. Save the file using :wq
14. Make executable using chmod +x
15. Execute using ./script.sh

SHELL SCRIPT CODE:

```
#!/bin/bash
# Name: A K

echo "---- START ----"

# Task 1: Display current directory
echo "Current Directory:"
pwd

# Task 2: List files
echo "Files in directory:"
ls

# Task 3: Create folder and file
echo "Creating folder and file..."
mkdir myfolder
touch myfolder/file1.txt

# Task 4: Word count of file
echo "Enter file name:"
read f
if [ -f "$f" ]
then
    wc -w "$f"
else
    echo "File does not exist"
fi

# Task 5: Reverse a string
echo "Enter text:"
read t
echo "Reversed string:"
echo "$t" | rev

# Task 6: Even or Odd
echo "Enter number:"
read n
if [  $$(n \% 2)$  -eq 0 ]
then
    echo "Even"
else
    echo "Odd"
fi

# Task 7: Largest of 3 numbers
echo "Enter three numbers:"
read a b c
big=$a

if [ $b -gt $big ]
then
    big=$b
fi

if [ $c -gt $big ]
then
    big=$c
fi

echo "Largest number is: $big"

# Task 8: Multiplication table
"script.sh" 96L, 1372B
```

```

read num
i=1
while [ $i -le 10 ]
do
    echo "$num x $i = $((num*i))"
    i=$((i+1))
done

# Task 9: Palindrome check
echo "Enter a word:"
read w
rev_word=$(echo "$w" | rev)

if [ "$w" = "$rev_word" ]
then
    echo "Palindrome"
else
    echo "Not a palindrome"
fi

# Task 10: Count files in directory
echo "Enter directory name:"
read dir

if [ -d "$dir" ]
then
    count=$(ls "$dir" | wc -l)
    echo "Total files in $dir: $count"
else
    echo "Directory does not exist"
fi

echo "--- END ----"

```

EXECUTION STEPS:

```

clover_99@LAPTOP-J302E0RV:/home$ vi script.sh
clover_99@LAPTOP-J302E0RV:/home$ pwd
/home
clover_99@LAPTOP-J302E0RV:/home$ ls
clover_99
clover_99@LAPTOP-J302E0RV:/home$ cd ~
clover_99@LAPTOP-J302E0RV:~$ vi script.sh
clover_99@LAPTOP-J302E0RV:~$ chmod +x script.sh
clover_99@LAPTOP-J302E0RV:~$ ./script.sh

```

OUTPUT:

```

---- START ----
Current Directory:
/home/clover_99
Files in directory:
script.sh
Creating folder and file...
Enter file name:
File
File does not exist
Enter text:
Bankai!! Ronoroara
Reversed string:
araoronoR !!iaknaB
Enter number:
2344444
Even
Enter three numbers:
324 4234 423
Largest number is: 4234
Enter number for table:
17
17 x 1 = 17
17 x 2 = 34
17 x 3 = 51
17 x 4 = 68
17 x 5 = 85
17 x 6 = 102
17 x 7 = 119
17 x 8 = 136
17 x 9 = 153
17 x 10 = 170
Enter a word:
MAAM
Palindrome
Enter directory name:
mkdir sample
touch sample/a.txt sample/b.txt sample/c.txtDirectory does not exist
--- END ---

```

CONCLUSION:

The shell script was successfully created using vi editor and executed using chmod +x. Various operations like file handling, loops, and conditional statements were performed, helping in understanding shell scripting concepts.